

CSCI 160: Computer Architecture I

Summer 2026 Syllabus

Instructor: Roy Delgado

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Meeting Time: Monday – Thursday, 5:45 PM – 7:15 PM

Room: HN-1001E

Office Hours: By request via email or in person

Prerequisites:

CSCI 12700 and CSCI 15000 with a passing grade.

Course Description:

This course introduces the fundamental concepts of computer architecture and digital logic design as used in modern computing systems. Students will learn how digital circuits are designed and how hardware components interact to execute software instructions. Topics include binary numbers, Boolean algebra, logic gates, combinational and sequential circuits, registers, counters, and memory systems. The course emphasizes the relationship between hardware and software and builds a foundation for understanding how computers operate at a low level. This course is a core requirement for the Computer Science major. Students are expected to successfully complete the course with a grade of C or higher.

Course Objectives:

- Understand and apply binary number systems and Boolean algebra.
- Design and analyze combinational circuits using logic gates.
- Implement and evaluate sequential circuits including flip-flops, registers, and counters.
- Explain how memory systems function within a computer.
- Demonstrate the relationship between hardware components and software execution.

Communication:

Email is the primary method of communication. Please include “CSCI 160” in the subject line of all emails. I will try to respond as fast as I can. I also answer emails during weekends. If you would like to discuss your course standing or performance, please attend office hours. Students are expected to regularly check their Hunter email and the course webpage for important course announcements and updates.

Required Materials:

The required textbook for this course is *Digital Design, 5th Edition* by M. Morris Mano and Michael D. Ciletti (Pearson Prentice Hall, 2012, ISBN-10: 0132774208).

All students should have access to course website, where lecture notes and additional course materials will be posted. In addition, each student must have a Gradescope account. I will send out an invitation to join the class to the emails I have on file.

Grading:

The grade for this course will be determined according to the following components:

Assessment	Percentage
Homework Assignments	20%
Midterm 1	20%
Midterm 2	20%
Final Exam	40%
Total	100%

Homework:

Homework will be assigned weekly and will consist of problems based on the material covered in class. Assignments will typically be uploaded at the start of class on the Mondays that they are assigned and will be due by the following Monday at 10:00 AM. There will be a total of five homework assignments. The lowest homework grade will be dropped. All homework must be submitted on Gradescope by the specified deadline. No late submissions will be accepted.

Midterm Exams:

There will be two **in-person** midterm exams during the semester. Each midterm will count for 20% of the overall course grade. Students are expected to come prepared with a pen or pencil. No calculators are needed for this course. An additional page of scrap paper will be provided for each exam. If you miss a midterm exam without a valid excuse, you will receive a score of zero. If you miss a midterm exam with a valid excuse, the grade you receive on the final exam will replace the missed midterm exam grade.

Final Exam:

The final exam will take place on the last day of class, July 8th. The exam will not be cumulative and will instead cover the material presented after Midterm 2. However, it is important to understand earlier course content, as concepts in this course build on one another. Additional details regarding the final exam will be provided later in the semester. If you miss the final exam, you will receive a WU for the course.

Extra Credit:

There will be several **in-person** exercises offered as extra credit throughout the semester. These exercises will be given without prior notification and will be available only to **students who attend class**. Each exercise **will count for 1% extra credit**. Students may accumulate up to a maximum of 5% extra credit over the course of the semester. There will be no make-up opportunities for these exercises.

Policy on Sexual Misconduct:

In compliance with the CUNY Policy on Sexual Misconduct, Hunter College reaffirms the prohibition of any sexual misconduct, which includes sexual violence, sexual harassment, and gender-based harassment retaliation against students, employees, or visitors, as well as certain intimate relationships. Students who have experienced any form of sexual violence on or off campus are entitled to the rights outlined in the Bill of Rights for Hunter College.

Sexual Violence: Students are strongly encouraged to immediately report the incident by calling 911, contacting NYPD Special Victims Division Hotline (646-610-7272), their local police precinct, or contacting the College's Public Safety Office (212-772-4444).

All Other Forms of Sexual Misconduct: Students are encouraged to contact the College's Title IX Campus Coordinator, Dean John Rose (jtrose@hunter.cuny.edu, 212-650-3262) or Colleen Barry (colleen.barry@hunter.cuny.edu, 212-772-4534).

Counseling Services: Students may seek complimentary services through the Counseling and Wellness Services Office, Hunter East 1123.

Academic Integrity:

Cheating: The academic integrity policy applies fully to this course. Students are encouraged to work together by forming study groups, discussing course material, and helping one another understand concepts. However, collaboration is strictly prohibited on exams. Any violation will be considered cheating and will result, at a minimum, in a failing grade for the course. All submissions must represent your own work and effort. Submitting work completed in part or whole by another individual constitutes academic dishonesty. It is also your responsibility to ensure that all submitted work runs correctly and is accessible to the instructor. Hunter College regards acts of academic dishonesty including plagiarism, cheating on examinations, obtaining unfair advan-

tage, and falsification of records as serious offenses and will enforce the CUNY Policy on Academic Integrity accordingly.

ADA Compliance:

Qualified students with disabilities will be provided reasonable academic accommodations if determined eligible by the Office of AccessABILITY. Prior to receiving accommodations in this course, the instructor must receive written verification of eligibility. In compliance with the Americans with Disabilities Act (ADA) and Section 504 of the Rehabilitation Act, Hunter College is committed to ensuring equal access and accommodations for students with documented disabilities and/or medical conditions. The College does not discriminate on the basis of pregnancy or related conditions. Students with documented disabilities emotional, medical, physical, and/or learning are encouraged to contact the Office of AccessABILITY to secure appropriate accommodations.

Course Schedule:

Date	Topic	Reading	Assignment
May 28th	Introductions	Syllabus	
June 1st	Digital Systems and Binary	1.1, 1.2	HW1 Assigned
June 2nd	Number-base Conversions	1.3, 1.4	
June 3rd	Complements and Signed Binary	1.5, 1.6	
June 4th	Binary Codes and Logic	1.7, 1.9	
June 8th	Boolean Algebra	2.1, 2.2, 2.3	HW1 Due/HW2 Assigned
June 9th	Theorems in Boolean Algebra	2.4, 2.5	
June 10th	Canonical and Standard Form	2.6	
June 11th	Logic Gates and Integrated Circuits	2.7, 2.8, 2.9	
June 15th	Midterm 1		HW2 Due/HW3 Assigned
June 16th	Karnaugh Maps	3.1, 3.2	
June 17th	K-Maps with Four Variables	3.3	
June 18th	POS + Don't-Care Conditions	3.4, 3.5	
June 22nd	No Class		HW3 Due/HW4 Assigned
June 23rd	NAND/NOR Implementation	3.6, 3.7, 3.8	
June 24th	Circuit Analysis and Design	4.1, 4.2	
June 25th	Circuit Analysis and Design	4.3, 4.4	
June 29th	Adders and Subtractors	4.5	HW4 Due/HW5 Assigned
June 30th	Midterm 2		
July 1st	Decimal Adders	4.6, 4.8	
July 2nd	Decoders and Encoders	4.9, 4.10	
July 6th	Multiplexers	4.11, 4.12	HW5 Due
July 7th	Intro to Sequential Logic	5.2	
July 8th	Final Exam		